



LGL3: Pathology of non-neoplastic intestinal diseases

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Inflammatory Bowel Disease (IBD)

- Chronic conditions resulting from inappropriate mucosal immune activation.
- Characteristics: chronic remitting relapsing inflammatory condition of unknown etiology.
- IBD encompasses two major entities:
 - **Crohn disease (CD)**
 - **Ulcerative colitis (UC)**

■ **Pathogenesis:**

- Remains uncertain.
- However, most investigators believe that IBD results from a combination of:-
 - Alteration in host reaction to intestinal microbiota.
 - Intestinal epithelial dysfunction.
 - Aberrant mucosal immune responses.
 - Altered composition of the gut microbiome.

- **Clinical features:**

- Alteration of bowel habits.
- Rectal bleeding, urgency and tenesmus.
- Abdominal cramping and pain.
- Nausea and vomiting.
- Systemic symptoms include: weight loss, fever, sweating, malaise, and arthralgia.

Extra intestinal manifestations of IBD:



Eyes
Episcleritis
Uveitis



Kidneys
Stones
(nephrolithiasis)
Hydronephrosis
Fistulae
Urinary tract
infection

Skin
Erythema nodosum
Pyoderma
gangrenosum



MS
GHE

Mouth
Stomatitis
Aphthous ulcers

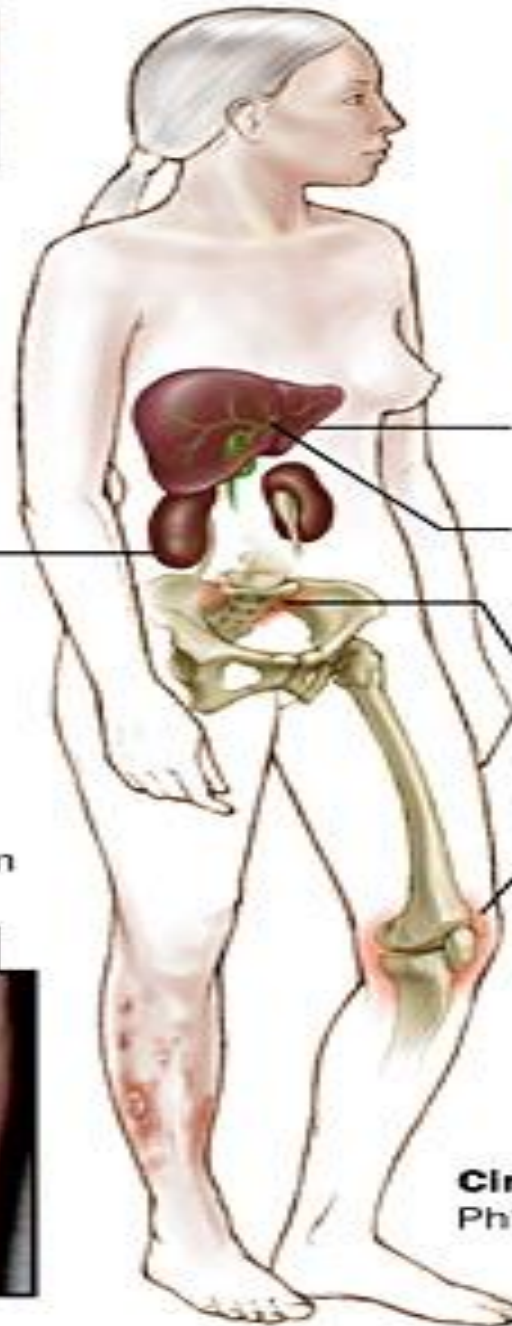


Liver
Steatosis

Biliary tract
Gallstones
Sclerosing cholangitis

Joints
Spondylitis
Sacroiliitis
Peripheral arthritis

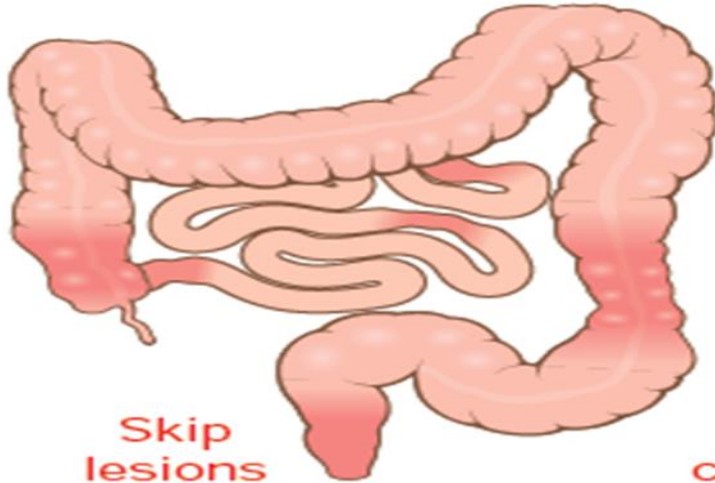
Circulation
Phlebitis



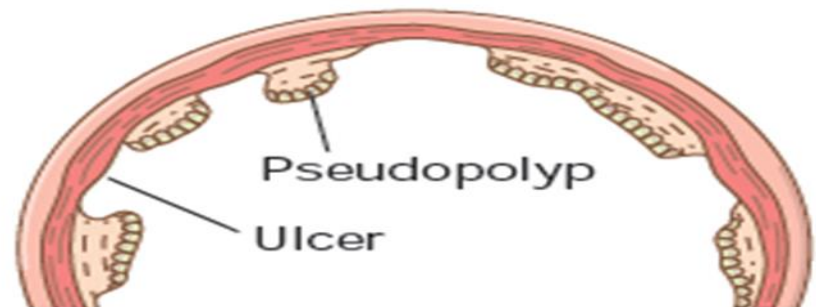
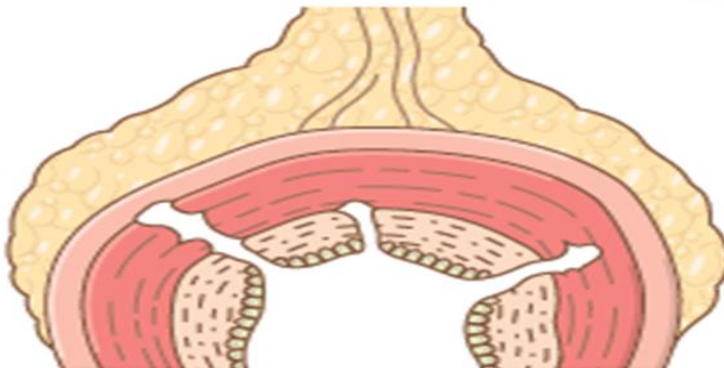
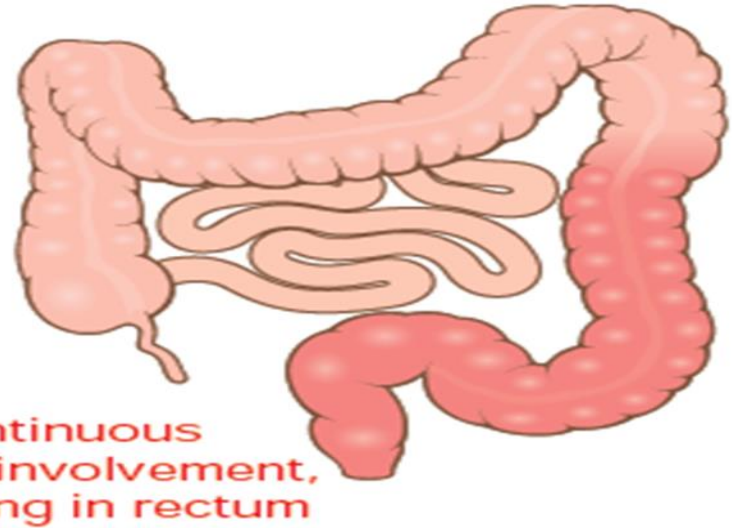
Clinical findings	CD	UC
Site of involvement	Any part of the alimentary tract from esophagus to anus	Rectum and colon
Malabsorption	Yes	No
Toxic megacolon	No	Yes
Perianal fistula	Yes	No
Recurrence after Surgery	Common	No
Malignant potential	Less	more
Smoking	Higher risk	Lower risk
Extraintestinal manifestations	Less	more

Feature	Crohn Disease	Ulcerative Colitis
Macroscopic		
Bowel region	Ileum ± colon	Colon only
Distribution	Skip lesions	Diffuse
Stricture	Yes	Rare
Wall appearance	Thick	Thin
Microscopic		
Inflammation	Transmural	Limited to mucosa
Pseudopolyps	Moderate	Marked
Ulcers	Deep, knife-like	Superficial, broad-based
Lymphoid reaction	Marked	Moderate
Fibrosis	Marked	Mild to none
Serositis	Marked	Mild to none
Granulomas	Yes (~35%)	No
Fistulae/sinuses	Yes	No

CROHN DISEASE

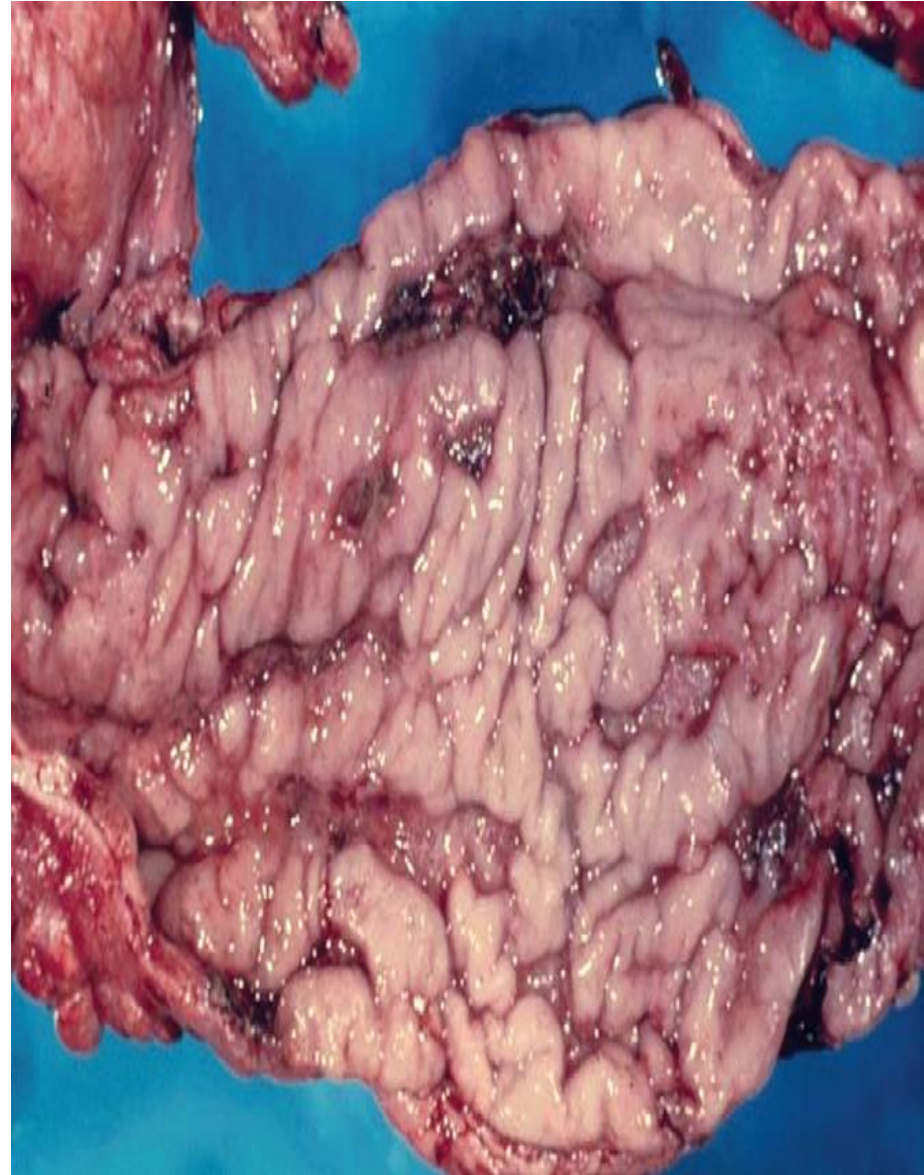
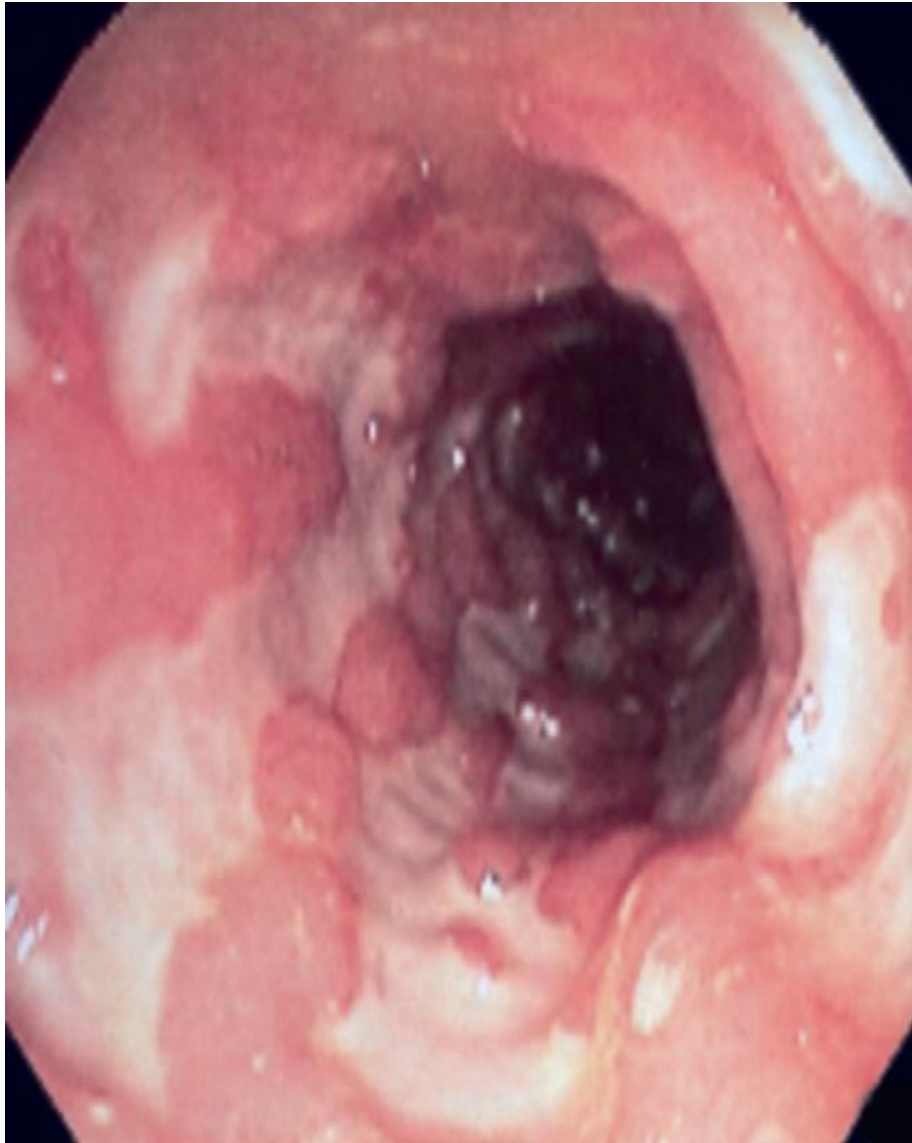


ULCERATIVE COLITIS



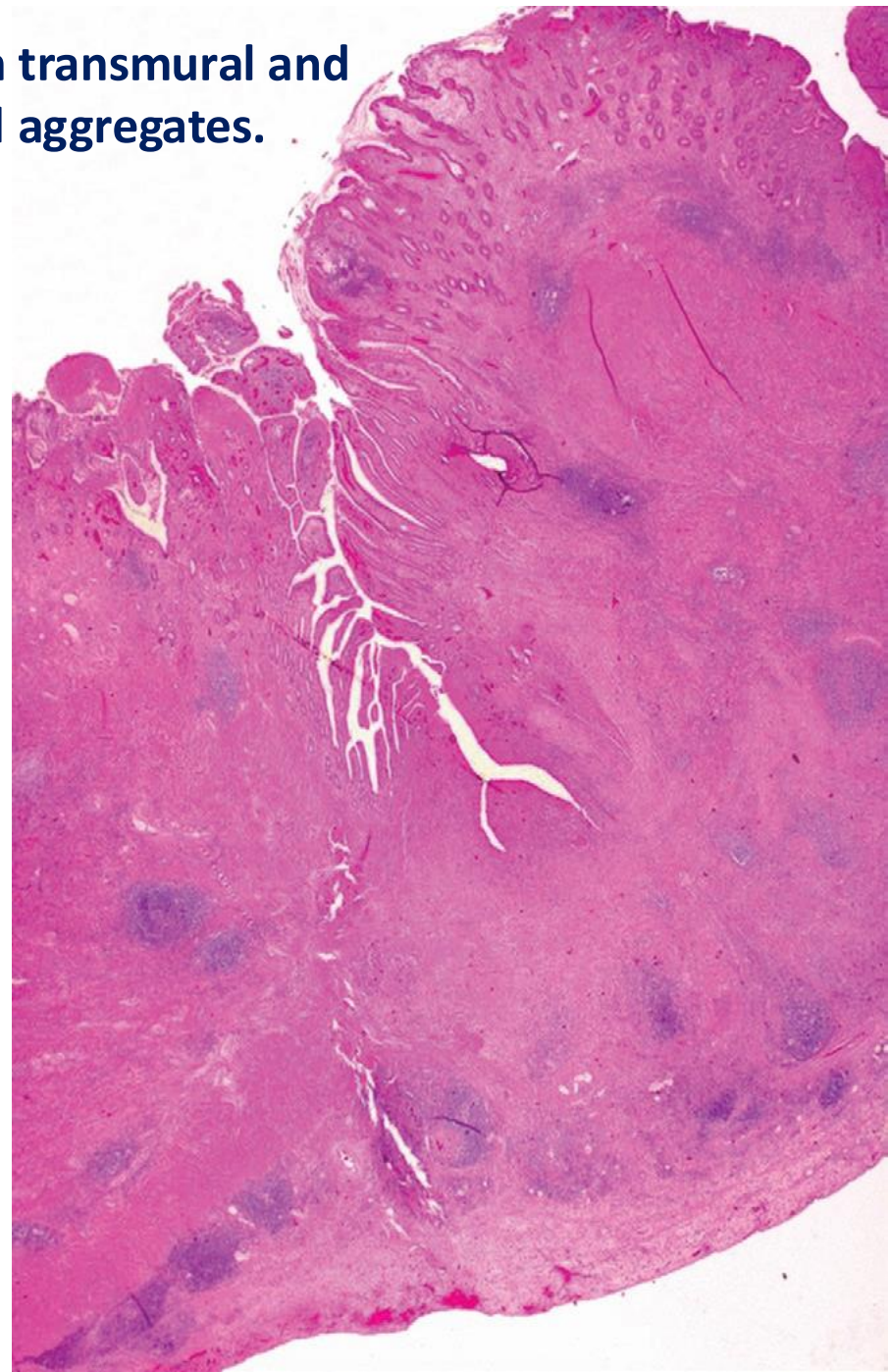
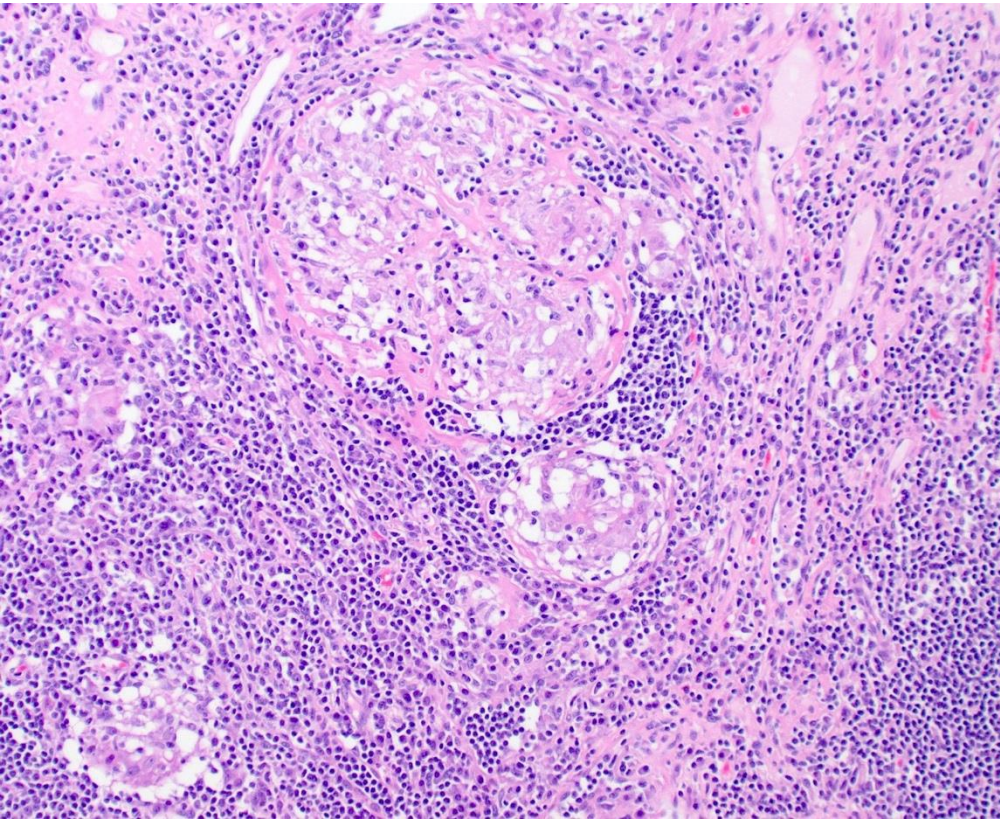
Distribution of lesions in inflammatory bowel disease. The distinction between Crohn disease and Ulcerative colitis is based primarily on morphology.

Crohn disease: longitudinal ulcers and intervening normal mucosa.



Crohn disease: Marked fibrosis with transmural and subserosal lymphoid aggregates. Fissuring ulceration.

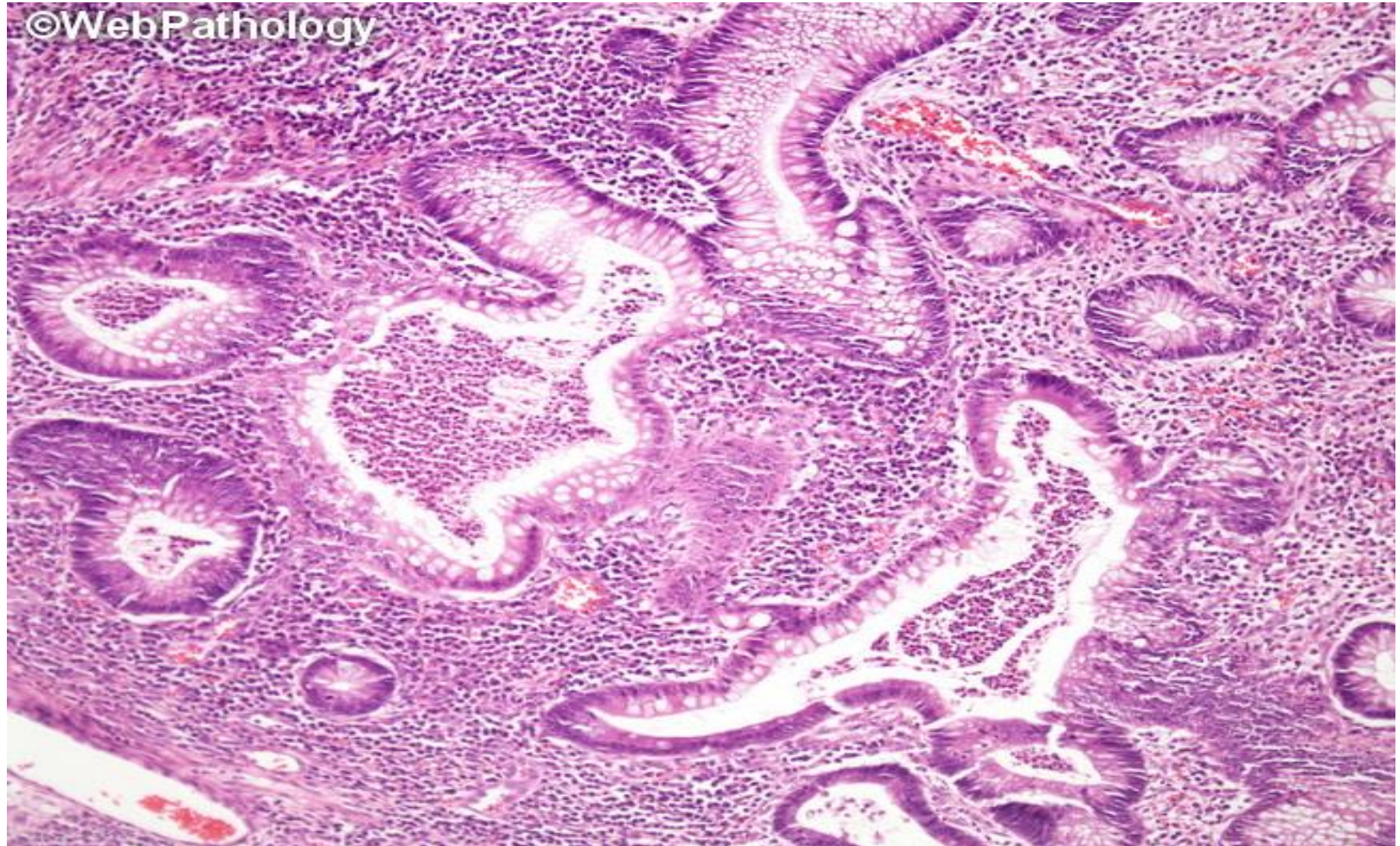
Non-necrotizing granuloma



Ulcerative colitis (UC): A colectomy specimen with UC:
Note the continuous involvement of the right half of the specimen and abrupt transition to normal mucosa. The affected mucosa is hyperemic and shows numerous pseudopolyps.



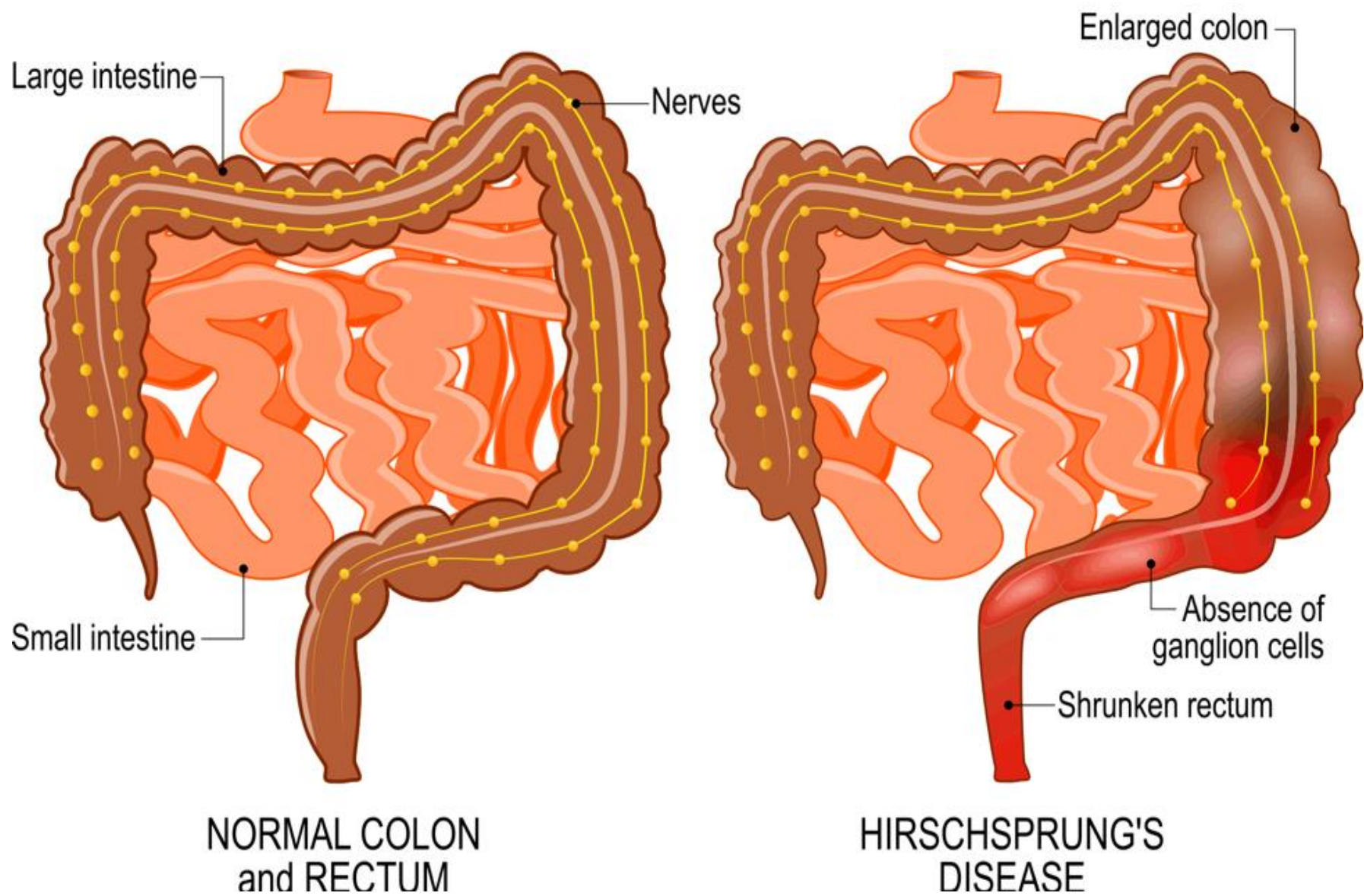
Biopsy from a patient with **Ulcerative colitis** showing crypt abscesses, marked inflammation and crypt distortion. Some of the glands show regenerative atypia with mucin depletion .



Hirschsprung disease

- Also known as **congenital aganglionic megacolon**
- It occurs in about 1 in 5000 live births.
- It presents with a failure to pass meconium in the immediate postnatal period, and may progress to abdominal distention and bilious vomiting.

- Results when the normal migration of neural crest cells from cecum to rectum is arrested prematurely; or when the ganglion cells undergo premature death.
- This produces a distal intestinal segment that lacks both the Meissner submucosal plexus, and the Auerbach (myenteric) plexus (“aganglionosis”).
- This results in absence of peristalsis and functional obstruction.

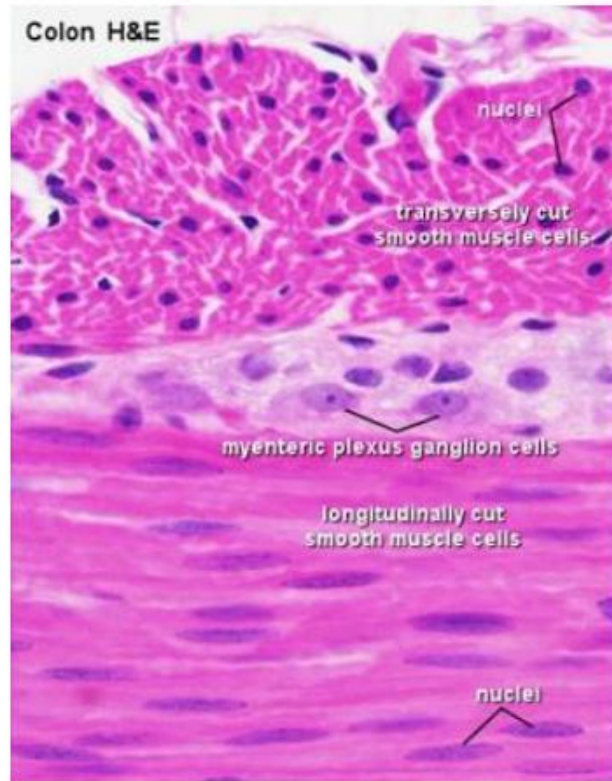


Contrast Enema:

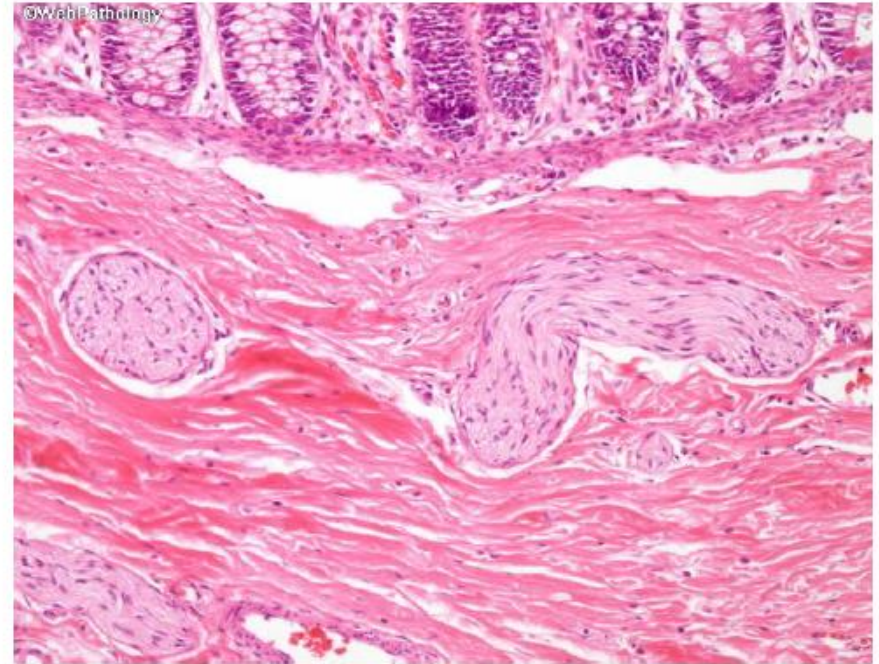
The affected segment is of small caliber with proximal dilatation of normal part.



Normal ganglionic cells



Aganglionic colon



Diagnosis: rectal biopsy shows absence of ganglion cells within the affected segment with hypertrophied nerve fibers.

- **Complications:**

- Enterocolitis
- Fluid and electrolyte disturbances.
- Perforation and peritonitis.

- **Treatment:** surgical resection of the aganglionic segment followed by anastomosis.

References

- **Robbins & Cotran Pathologic Basis of Disease - 9th edition -2015.**
- **Robbins Basic Pathology - 10th edition 2018.**
- **Muir's Textbook of Pathology 15th edition – 2014.**
- **Rubin's Pathology, Clinicopathologic Foundations of Medicine – 6th edition -2012.**
- **WEBSITE**